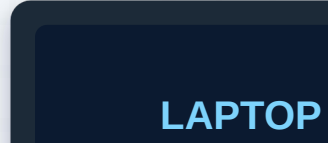


56 numbered wires · every hole named (column 1–63, row a–j) · pins match rig_firmware_v2.ino · wire with EVERYTHING unplugged

```
I2C: A4 = SDA · A5 = SCL · multiplexer 0x70
```



go in order: power → drivers → STEP/DIR → motors → sensor → laser → vibration → USB

- 39 multiplexer SDA → Nano A4 [j11]
- 40 multiplexer SCL → Nano A5 [j12]
- 41 multiplexer A0 → BOTTOM blue rail (GND) [col 13]
- 42 multiplexer A1 → BOTTOM blue rail (GND) [col 16]
- 43 multiplexer A2 → BOTTOM blue rail (GND) [col 18]
- 44 multiplexer SD0 → AS5600 SDA
- 45 multiplexer SC0 → AS5600 SCL
- 46 AS5600 VCC → BOTTOM red rail (+5 V) [col 51]
- 47 AS5600 GND → BOTTOM blue rail (GND) [col 52]
- 48 AS5600 DIR (if your board has it) → BOTTOM blue rail (GND) [col 53]
- 49 laser S → Nano D7 [a9]
- 50 laser → TOP blue rail (GND) [col 7]
- 51 Nano D8 [c8] → hole b20 (top of resistor R1) [b20]
- 52 transistor E (emitter) [g19] → BOTTOM blue rail (GND) [col 19]
- 53 diode stripe + motor red strip [g24] → BOTTOM red rail (+5 V) [col 24]
- 54 vibration motor red (+) → hole g24 (+5 V strip) [g24]
- 55 vibration motor blue / black (−) → hole g21 (transistor C) [g21]
- 56 Nano USB-C → laptop (through the hub)

C1	100 μ F at col 25: long leg + (red), stripe T- (blue)
C2	100 μ F at col 37: same way round
J	part: short jumper j31 .. j32 (RST-SLP)
J	tilt: short jumper j43 .. j44
R1	1 k Ω , E20 .. f20 across the gap (brown-black-red)
Q1	2N2222 legs in j19 j20 j21, flat face toward you (E B C)
D1	1N4148, f21 .. f24, black stripe at f24
V	ref 0.55 V on both drivers (12 V OFF)

next to each rail row on YOUR board.

Beep-test: TOP red ↔ BOTTOM red = SILENT.

12 V must never reach 5 V, the Nano or VDD.

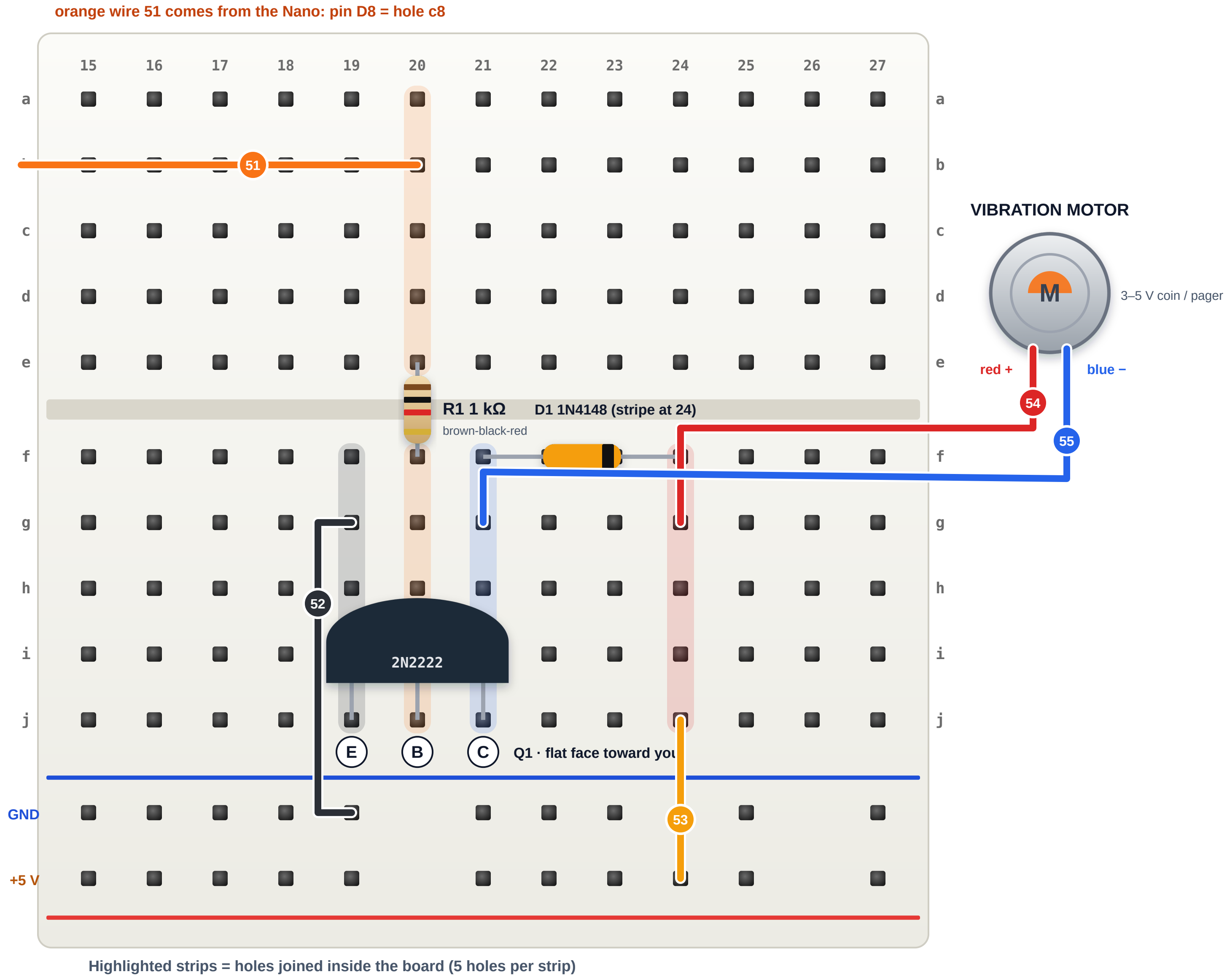
Power ON: USB, then 12 V switch.

Power OFF: 12 V switch, then USB.

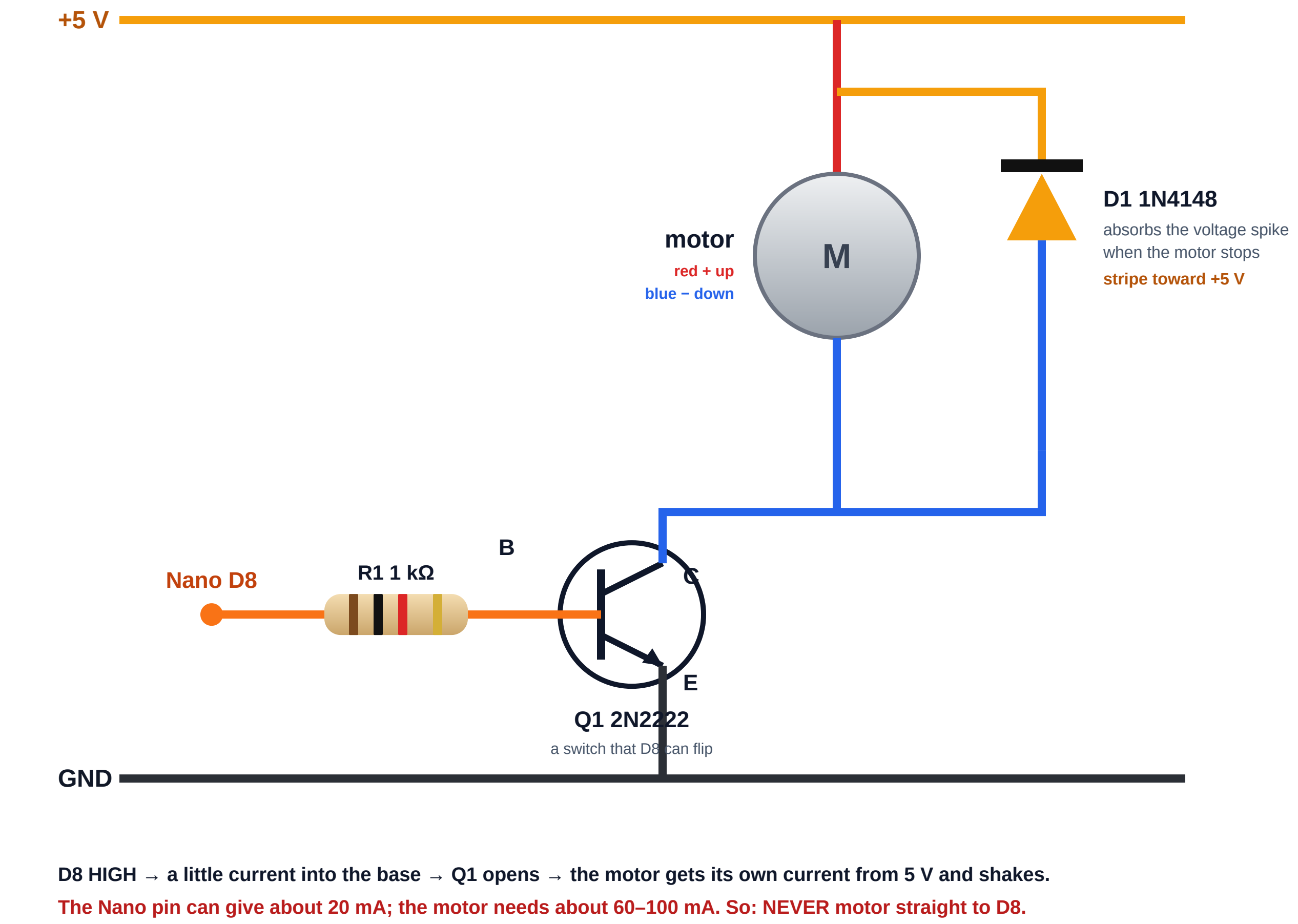
ZeroDrift MK2 · vibration motor

the “platform shake” demo: serial V1 = shake ON · V0 = OFF · Arduino pin D8 drives it through a transistor, never straight from the pin

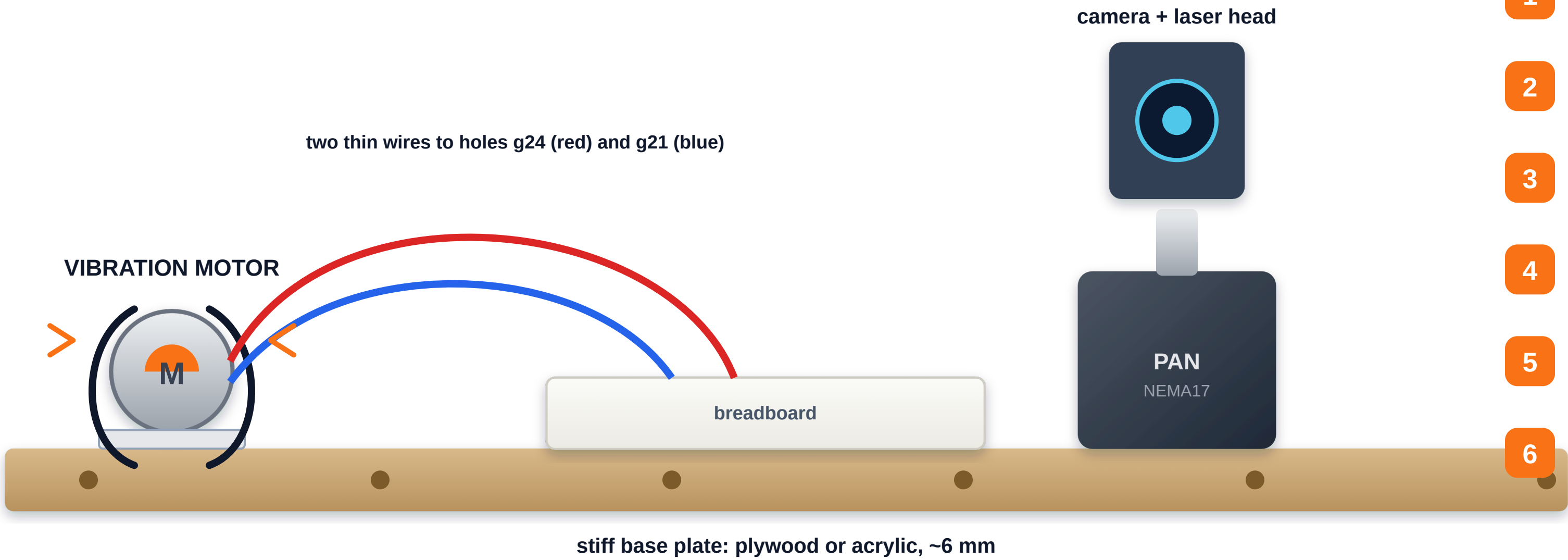
1 On the breadboard: columns 15 – 27



2 The circuit (why a transistor)



3 Where the vibration motor is fixed on the rig



How to fix it

- 1 Put it on the BASE PLATE, 10–15 cm from the pan motor. The plate carries the whole rig, so the whole rig shakes.
- 2 Fix it flat with a cable tie AND a strip of double-sided foam tape. Bolt only if it comes with a mounting ear.
- 3 Wires: 20–30 cm, run along the plate edge with tape. Keep them away from the pan shaft and the AS5600 magnet.
- 4 Plug the red wire into g24 and the other wire into g21 (close-up, panel 1).
- 5 Test with 12 V OFF: in the serial monitor type V1 → you feel a buzz, V0 → it stops.
- 6 Demo moment: rig locked on the beacon → V1 → the lock box shakes but holds → V0.

ZEROdrift · MK2 · connection diagram

every connection of the breadboard sheet, drawn as a schematic · a tag (5V / GND) means: wire that pin to that rail

